PROBLEM 3 · WHITEBOARD GUIDE

The plate with four interior nodes, solved by hand

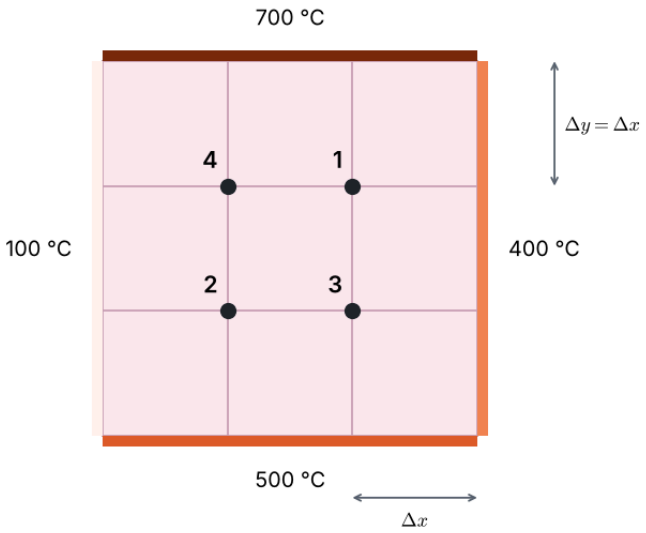
A board-by-board guide for teaching the Finite Difference Method with a pocket calculator, before showing the code

Aim of the lesson. With four unknowns, this problem is the smallest possible two-dimensional finite difference problem, which makes it ideal for the board: every number can be computed with a pocket calculator in a few minutes. Students solve it **twice**, by **Gauss–Seidel iteration** (a table of averages) and by **direct elimination** (a few lines of algebra), then check the answer themselves. Every intermediate number is given below.

Lesson plan (about 45 minutes)

Board	Content	Time
1	The plate, its edge temperatures and the node numbers	5 min
2	Laplace equation → "each node is the average of its neighbours"	10 min
3	The four node equations	5 min
4	Method A: Gauss–Seidel iteration by hand	10 min
5	Method B: direct elimination with sums and differences	8 min
6	Checks: averages, energy balance, exact solution	7 min

Board 1 · The plate



The square plate divided into 3 × 3 squares, its edge temperatures and the four interior points.

Copy the figure carefully, **with the node numbers exactly as in the problem**: 4 is top-left, 1 is top-right, 2 is bottom-left, 3 is bottom-right.

Ask the class: *Where do you expect the hottest interior point? The coldest?* (Node 1, next to the 700 °C and 400 °C edges; node 2, next to the 100 °C edge.)

Board 2 · From the Laplace equation to averages

Steady conduction, no heat generation, constant k:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0.$$

Central differences at node P, with neighbours East, West, North, South:

$$\frac{T_E - 2T_P + T_W}{\Delta x^2} + \frac{T_N - 2T_P + T_S}{\Delta y^2} = 0.$$

With $\Delta x = \Delta y$, multiply by Δx^2:

$$T_E + T_W + T_N + T_S - 4T_P = 0 \qquad \Longleftrightarrow \qquad \boxed{T_P = \frac{T_E + T_W + T_N + T_S}{4}}$$

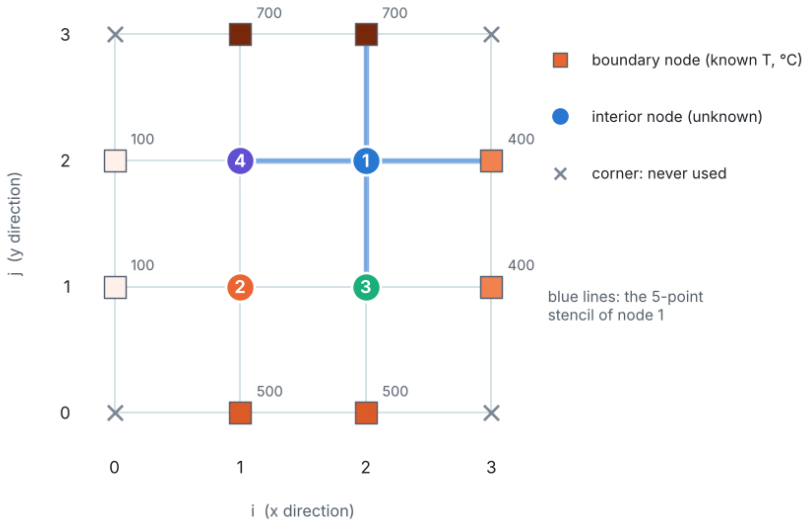
Every interior temperature is the average of its four neighbours.

Ask the class: *Why do the plate size, the grid spacing and the conductivity not appear? Do we need the corner temperatures?* (No: the formula only uses the four side neighbours.)

Board 3 · The four equations

Fill in this table on the board together, one node at a time:

node	East	West	North	South	equation
1	400	T_4	700	T_3	$4T_1 = 1100 + T_3 + T_4$
2	T_3	100	T_4	500	$4T_2 = 600 + T_3 + T_4$
3	400	T_2	T_1	500	$4T_3 = 900 + T_1 + T_2$
4	T_1	100	700	T_2	$4T_4 = 800 + T_1 + T_2$



The grid with known edge nodes, the four unknowns and the five-point stencil of node 1.

Board 4 · Method A: Gauss–Seidel by hand

Use each equation as an update rule, and always use the **newest** values:

$$T_1 \leftarrow \frac{1100 + T_3 + T_4}{4}, \quad T_2 \leftarrow \frac{600 + T_3 + T_4}{4}, \quad T_3 \leftarrow \frac{900 + T_1 + T_2}{4}, \quad T_4 \leftarrow \frac{800 + T_1 + T_2}{4}.$$

> **Calculator shortcut.** T_1 and T_2 both need $S = T_3 + T_4$, and T_3 and T_4 both need $T_1 + T_2$: compute each sum once per iteration.

Start with a guess of **400 °C** at all four nodes. **Iteration 1**, worked out on the board:

$$T_1 = (1100 + 400 + 400)/4 = 475.00$$
$$T_2 = (600 + 400 + 400)/4 = 350.00$$
$$T_3 = (900 + 475 + 350)/4 = 431.25 \quad (\text{uses the new } T_1, T_2)$$
$$T_4 = (800 + 475 + 350)/4 = 406.25$$

Let the students do iteration 2 themselves, then complete the table:

iteration	T_1	T_2	T_3	T_4	largest change
0 (guess)	400.00	400.00	400.00	400.00	
1	475.00	350.00	431.25	406.25	75.0
2	484.38	359.38	435.94	410.94	9.38
3	486.72	361.72	437.11	412.11	2.34
4	487.30	362.30	437.40	412.40	0.59
5	487.45	362.45	437.48	412.48	0.15
6	487.49	362.49	437.49	412.49	0.04

Point out: after the first iteration, the largest change is divided by **4** every time (9.38 → 2.34 → 0.59 → 0.15 → 0.04). Six iterations give the answer to better than 0.1 °C.

Board 5 · Method B: direct elimination with sums and differences

Rewrite the equations with all unknowns on the left:

$$(1) \quad 4T_1 - T_3 - T_4 = 1100 \qquad (3) \quad 4T_3 - T_1 - T_2 = 900$$
$$(2) \quad 4T_2 - T_3 - T_4 = 600 \qquad (4) \quad 4T_4 - T_1 - T_2 = 800$$

Differences:

$$(1) - (2):\ 4(T_1 - T_2) = 500 \ \Rightarrow\ T_1 - T_2 = 125, \qquad (3) - (4):\ 4(T_3 - T_4) = 100 \ \Rightarrow\ T_3 - T_4 = 25.$$

Sums, with $S = T_1 + T_2$ and $Q = T_3 + T_4$:

$$(1) + (2):\ 4S - 2Q = 1700, \qquad (3) + (4):\ 4Q - 2S = 1700.$$

Subtracting gives $6S - 6Q = 0$, so $S = Q$, and then $2S = 1700$: $S = Q = 850$.

Answer:

$$T_1 = \frac{850 + 125}{2} = 487.5, \quad T_2 = \frac{850 - 125}{2} = 362.5, \quad T_3 = \frac{850 + 25}{2} = 437.5, \quad T_4 = \frac{850 - 25}{2} = 412.5\ ^\circ\text{C}.$$

Both methods agree. (The notebook also solves the same system by standard Gaussian elimination, printing the matrix after every step.)

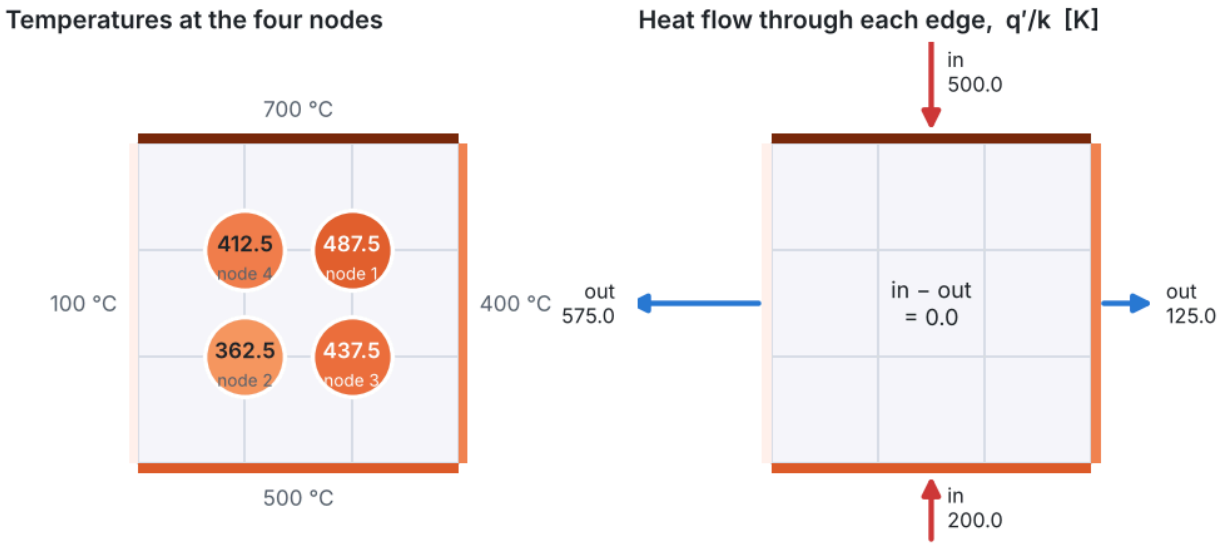
Board 6 · Checks the class can do by hand

1. Every node is the average of its neighbours:

node	neighbours	average
1	$(400 + 700 + 412.5 + 437.5)/4$	487.5 ✓
2	$(100 + 500 + 412.5 + 437.5)/4$	362.5 ✓
3	$(400 + 500 + 487.5 + 362.5)/4$	437.5 ✓
4	$(100 + 700 + 487.5 + 362.5)/4$	412.5 ✓

2. Energy balance (heat in = heat out). Between an edge and its interior neighbour the heat flow is $q'/k = T_{edge} - T_{node}$ (with $\Delta x = \Delta y$):

edge	q'/k [K]	
top (700 °C)	$(700-412.5)+(700-487.5)=+500$	in
bottom (500 °C)	$(500-362.5)+(500-437.5)=+200$	in
left (100 °C)	$(100-412.5)+(100-362.5)=-575$	out
right (400 °C)	$(400-487.5)+(400-437.5)=-125$	out
sum	**0**	✓



The four temperatures on the plate and the heat flows through the edges.

3. Comparison with the exact solution (series solution, computed in the notebook):

node	1	2	3	4
by hand, 4 nodes [°C]	487.5	362.5	437.5	412.5
exact [°C]	490.4	359.6	438.1	411.9
difference [°C]	−2.9	+2.9	−0.6	+0.6

> **Four equations and a pocket calculator give every temperature within 3 °C of the exact solution.** The notebook shows that finer grids close this gap, the error falling by about 4 each time the grid spacing is halved.

Questions for a discussion

- **Start Gauss–Seidel from 425 °C** (the average of the four edge temperatures). It converges in **one** iteration! Why? (Because $425+425=850=T_3+T_4$ exactly, so the first two updates are already exact, and so are the next two. See Board 5.)
- Use **Jacobi** instead (update all four nodes from the *old* values only). It still converges, but more slowly: the change is divided by 2, not 4, per iteration.
- How would the equations change if the bottom edge were **insulated**? (Nodes 2 and 3 get a ghost node below them, as in Problem 1.)

Common mistakes to watch for

- **Mixing up the node numbers**: 4 is top-left and 1 is top-right.
- Using the **corner** temperatures: they never appear in the five-point formula.
- Forgetting to use the **newest** values in Gauss–Seidel (that is Jacobi, which is slower).
- Dividing by 4 too early in the elimination and losing precision.

From the board to the notebook

`03_fdm_solver.ipynb` repeats every board: it **assembles** the same four equations automatically, solves them by Gaussian elimination and by Gauss–Seidel (from 400 °C, giving the table of Board 4), runs the energy check of Board 6, and then solves the same plate on grids of up to 96 × 96 squares, which nobody would want to do by hand.